Table 1: Accuracy and cost on AR-Bench, **held-out test split**, 100 episodes per cell, with a frozen Qwen2.5-7B-Instruct and a budget of 25 turns. Cost is mean input plus output tokens per episode. Every method shown is reported on all three tasks. Guess Number is included for completeness but is saturated by the harness search and is not evidence about reasoning (§**??**).

	Guess Number		Detective Cases		Situation Puzzles	
Method	Acc.	Tok/ep	Acc.	Tok/ep	Acc.	Tok/ep
Zero-shot	0.0	31,094	33.0	143,614	7.0	37,568
Few-shot	0.0	42,378	32.0	268,291	3.5	52,450
Few-shot with instruction	1.0	54,925	39.0	286,701	5.5	54,713
Proactive CoT	2.0	45,509	27.0	160,959	0.0	45,997
Tree of Thoughts[†]	6.0	21,196	34.0	244,318	9.0	11,360
Ask nothing (anchor)	0.0	226	36.0	847	10.5	215
Ours, cost-efficient	**100.0**	**7**	57.0	18,308	31.5	10,913
Ours, best	**100.0**	**7**	**63.0**	25,715	**32.5**	16,142

[†] Repaired, and AR-Bench implements ToT separately for each task, so the three cells are three modules: `gn/tot` iterates characters instead of candidate guesses and cannot run as published; `sp/dc` ToT never assigns the state evaluator Yao et al. specify, leaving the choice to a tie-break. Both are corrected here. The Guess Number cost is input tokens only — that record keeps the guess list and not the conversation — which understates it by roughly 1%. Uncertainty-of-Thoughts is held out of the seed set and is not reported.

Selection split. The frontier was chosen on a disjoint selection split, and the winners give some of that edge back: Detective Cases 80.0 there against 63.0 here, Situation Puzzles 41.0 against 32.5. The anchor moves by at most 1.5 points and the cost-efficient Detective Cases system by none, which is what selection on a held-out frontier should look like (§**??**). The Guess Number frontier holds a single system, so both **Ours** rows are the same one there.